

Alexandru Ilies

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PROFESSIONAL SUMMARY

Senior Software Engineer and AWS-certified Cloud Architect with 11 years of experience delivering production systems at scale. I build event-driven, serverless architectures on AWS, lead full-stack teams through infrastructure and deployment, and work as a forward deployment engineer close to customers and real-world delivery. I also design and orchestrate AI agent workflows using spec-driven development to turn ambiguous problems into testable software. Holds 4 AWS certifications, including Security Specialty and DevOps Professional.

TECHNICAL SKILLS

Programming Languages: TypeScript, JavaScript, Go, Java, Python, Swift

Frameworks & Runtime: Angular, React, NestJS, Node.js, Spring, Tailwind, iOS/WatchOS

Cloud & Infrastructure: AWS (ECS, Lambda, SQS, SNS, EventBridge, CloudWatch, RDS, DynamoDB), Terraform, CDK, CloudFormation, Docker

Databases: PostgreSQL, DynamoDB (Single/Multi-Table Design), MongoDB, Aurora, Redis

DevOps & Tooling: GitHub Actions, GitLab CI/CD, Docker, Nx, Git

Architecture & Leadership: Event-Driven Architecture, Serverless, Microservices, Technical Mentoring, System Design

EXPERIENCE

Lead Software Engineer

Jan. 2020 – Present

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Cluj, RO

- Architected and owned the complete cloud infrastructure for a crowd testing platform serving 35,000+ users, using 20+ AWS services including ECS, Lambda (30+), SQS, SNS, EventBridge, and CloudWatch.
- Led a team of up to 10 developers across frontend and backend, driving code quality through structured reviews, automated testing, and mentorship programs.
- Implemented event-driven, serverless architectures that improved system scalability and reduced operational costs through continuous optimization.
- Established CI/CD pipelines with GitHub Actions and GitLab CI, enabling multiple deployments per day with zero-downtime releases.
- Defined and enforced security best practices across all projects — WAF, IAM policies, secrets management — aligned with AWS Security Specialty certification.

Software Engineer

July. 2015 – Dec. 2019

msg.group

Cluj, RO

- Progressed from junior to mid-level engineer across multiple client projects, building full-stack web applications with Angular, TypeScript, and Node.js.
- Delivered full feature modules — ticketing, user rewards, and public-facing portals — for a German public sector platform serving citizens nationwide.
- Built first cloud-native solutions on AWS, gaining hands-on expertise that led to 4 AWS certifications.

KEY PROJECTS

Passbrains | Lead Developer & Cloud Architect | *AWS (ECS, Lambda, SQS, SNS, EventBridge, CDK), Angular, NestJS*

- Led the complete re-architecture of a public crowd testing platform serving 35,000+ users, from monolith to event-driven serverless.
- Designed and deployed 30+ Lambda functions orchestrated via EventBridge and SQS/SNS for asynchronous processing.
- Managed infrastructure as code with CDK and Terraform, enabling reproducible multi-environment deployments.
- Drove DevSecOps adoption — automated vulnerability scanning, secrets rotation, and compliance checks in CI/CD pipelines.

Dipko | Full-Stack Developer | *AWS, Angular, TypeScript, Node.js*

- Built a German public sector platform aggregating government institutions into a unified citizen-facing portal.
- Deployed cloud-based solutions on AWS tailored to public sector compliance and availability requirements.

Personal Tools | *AWS, React, TypeScript, Go, Java*

- Serverless reverse proxy and link shortener: l.dot-core.com

EDUCATION & CERTIFICATIONS

Technical University of Cluj-Napoca

Bachelor in Computer Science

Cluj, RO

Oct. 2012 – July 2016

AWS Security Specialty [Verify](#)

AWS Associate Developer [Verify](#)

AWS DevOps Professional [Verify](#)

AWS Associate Architect [Verify](#)